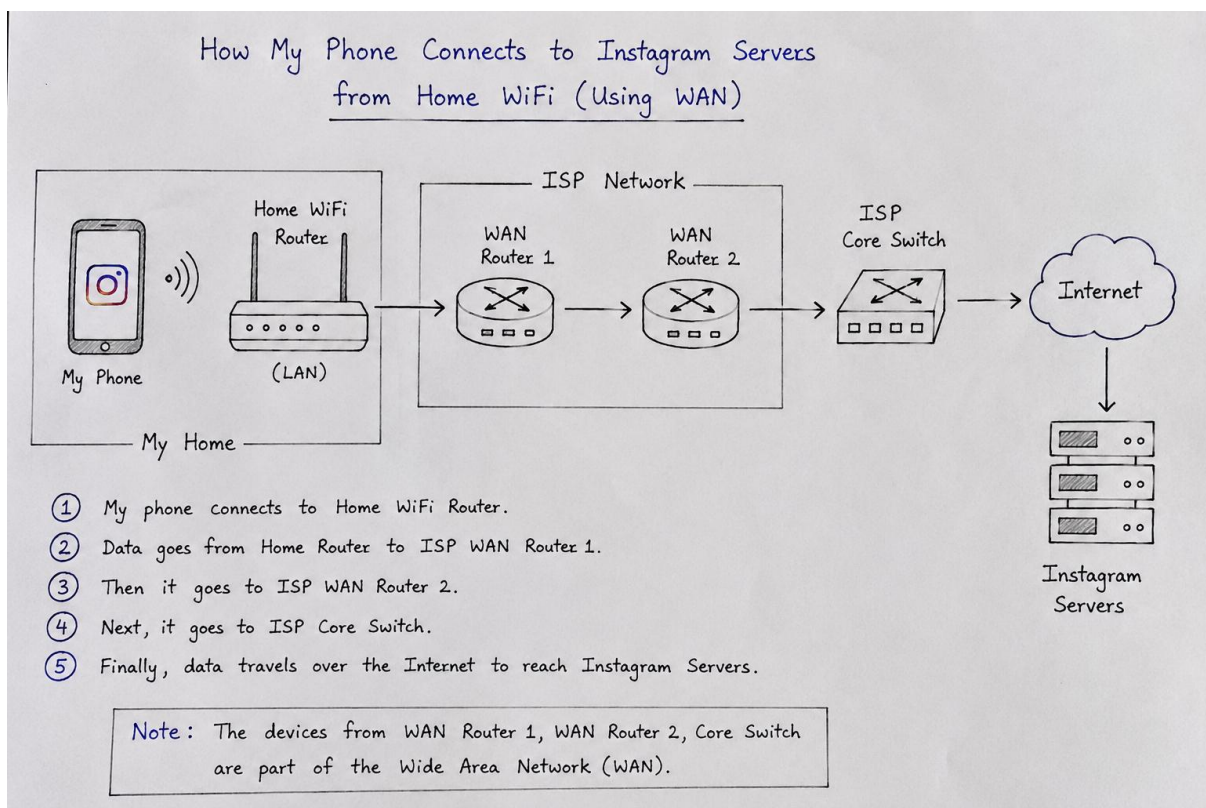


Wide Area Networks

Q1. Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.

Ans:

Figure 1: Communication from Home WiFi to Instagram Servers through WAN



Explanation

The phone first connects to the **Home WiFi Router**, which is part of the local network (LAN). The data is then forwarded to the **Internet Service Provider (ISP)**, where it passes through **WAN Router 1**, **WAN Router 2**, and an **ISP Core Switch** before reaching the Internet. Finally, the request reaches the **Instagram servers**, and the response follows the same path back to the phone.

WAN Devices Used

Device	Purpose
Home WiFi Router	Connects the phone to the internet.
WAN Router 1	Forwards data through the ISP network.
WAN Router 2	Routes data towards the destination network.
ISP Core Switch	Transfers data efficiently within the ISP network.

Conclusion

The diagram shows how data travels from a home WiFi network to Instagram servers through a Wide Area Network (WAN). WAN devices such as routers and switches help transfer data across long distances quickly and reliably.

Q2. Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).

Ans:

WAN Technology	Speed	Cost	Real-Life Usage
MPLS (Multiprotocol Label Switching)	High	High	Used by large companies such as Zomato offices , banks, and corporate branches for secure communication.
Leased Line	Very High (Dedicated Connection)	Very High	Used by IRCTC , data centres, hospitals, and organizations that require a reliable 24×7 internet connection.
Broadband Internet	Medium to High	Low	Used in homes , small offices, and shops for daily internet access.

Conclusion

Different WAN technologies are used based on business requirements. Large organizations prefer **MPLS** or **Leased Lines** for better speed, security, and reliability,

while **Broadband Internet** is the most common and affordable choice for homes and small businesses.

Q3. Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop. Hint: Focus on how the VPN creates a private tunnel over the public WAN.

Ans:

What is a VPN?

A **Virtual Private Network (VPN)** is a security technology that creates a secure and encrypted connection over a public **Wide Area Network (WAN)**, such as the Internet. It protects user data and privacy while accessing online services.

How a VPN Uses WAN

When you connect to a public WiFi network, such as one at a coffee shop, your internet traffic normally travels through the public WAN. Without protection, hackers on the same WiFi may try to intercept your data.

When a VPN is enabled, it creates an **encrypted tunnel** between your device and the VPN server. Instead of sending data directly to websites like **Spotify** or **YouTube**, your request first travels securely to the VPN server. The VPN server then forwards the request to the destination website and securely sends the response back through the same encrypted tunnel.

This process keeps your personal information private and makes it much more difficult for others to view or steal your data.

Real-Life Example

Scenario: Using YouTube at a Coffee Shop

1. Connect your phone to the coffee shop's public WiFi.
2. Turn on the VPN application.
3. The VPN creates a secure encrypted tunnel over the Internet (WAN).
4. Open YouTube.
5. Your request travels through the following path:

Phone → Coffee Shop WiFi → Internet (WAN) → VPN Server → YouTube Server

6. The response returns through the same encrypted VPN tunnel to your phone.

Advantages of Using a VPN

- Protects personal information on public WiFi.
- Encrypts internet traffic to improve security.
- Hides your real IP address.
- Reduces the risk of data theft.
- Allows safer access to online applications and websites.

Conclusion

A VPN uses the public WAN to create a secure and private encrypted tunnel between the user's device and the VPN server. This protects internet traffic, improves privacy, and allows users to safely access applications such as Spotify and YouTube, even when connected to public WiFi networks.

Q4. Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.

Ans:

For connecting Flipkart warehouses located in **Ahmedabad, Mumbai, and Delhi**, the best WAN technology is **MPLS (Multiprotocol Label Switching)**.

Reasons

1. Secure and Reliable Communication

MPLS provides a secure and reliable connection between different warehouse locations. This helps employees share inventory, order, and delivery information safely.

2. Fast Data Transfer Between Branches

MPLS offers low network delay and stable performance, allowing data to travel quickly between warehouses. This improves order processing and warehouse coordination. This helps the company manage its operations efficiently.